Student's Assessment number:..

PRIME MINISTER'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

KILIMANJARO REGIONAL COMMISSIONER'S OFFICE

FORM TWO MOCK ASSESSMENT

PHYSICS

CODE: 031
TIME: 2:30HRS

Wednesday, 20th May A.M 2026

INSTRUCTIONS

1. Write your answers **clearly and neatly** in the spaces provided/ Answer sheet given.
2. Use **blue or black ink** for writing. **Pencil should be used for diagrams only.**
3. Use **SI units** where applicable.
4. **Electronic devices are not allowed**, except a non-programmable calculator.
5. The constants used
   - The density of water, $\rho = 1000kg/m^3$.
   - Acceleration due to gravity, $g = 10ms^{-2}$

FOR ASSESSOR'S USE ONLY

QUESTION NUMBER	SCORE	INITIALS OF THE EXAMINER
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
TOTALS		
CHECKER'S INITIALS		

Student's Assessment number:...

SECTION A (15 marks)

Answer **all** questions

From the given alternatives, choose the letter of the most correct answer. **(1mark@)**

(i) A student notices that electric bulbs produce light when current flows, water boils when heated, and a ball changes direction when kicked. Which branch of physics best explains all these observations?

A. Physics is divided into unrelated branches
B. Physics studies matter, energy and their interactions
C. Physics deals only with motion
D. Physics only explains electrical phenomena

(ii) A student repeatedly measures the diameter of a wire and gets slightly different values each time. The best way to reduce this error is to:

A. Use a shorter ruler B. Measure only once
C. Take the average of several readings D. Ignore small differences

(iii) Two objects A and B have equal mass, but object A occupies more volume than B. What can be concluded?

A. A has higher density than B B. B has higher density than A
C. Both have the same density D. Density cannot be determined

(iv) Why are magnets often fitted to the doors of refrigerators and sum cupboards?

A. To keep away heat
B. To keep the inside environment warm
C. To keep the door tightly closed
D. To keep iron away

(v) An object immersed in water experiences an upthrust equal to:

A. Its mass B. Its volume
C. The weight of displaced liquid D. The density of the liquid

(vi) Why do sharp knives cut better than blunt ones?

A. They are heavier B. They have no mass
C. They are longer D. They have smaller area of contact

(vii) A suction cup sticks to a smooth wall because:

A. Air pressure outside is greater than inside
B. Gravity pulls it
C. The wall attracts it magnetically
D. It has glue

(viii) A car starts from rest and moves with constant acceleration. Which graph best represents this motion?

A. Distance–time straight line
B. Acceleration-time straight line with positive slope
C. Velocity–time horizontal line
D. Velocity–time straight line with positive slope

(ix) A camera forms an image on the film or sensor that is:

A. Real, inverted and diminished
B. Virtual, upright and magnified
C. Real, upright and magnified
D. Virtual, inverted and diminished

(x) A student lifts a 50 N load through a height of 2 m in 5 s. What is the power developed?

A. 10 W B. 20 W
C. 50 W D. 100 W

i	ii	iii	iv	v	vi	vii	viii	ix	x

Student's Assessment number:..

2. Match each item from **column A** with its corresponding item in **column B**. Then write letter of the correct response in the space provided. (1mark@)

COLUMN A	COLUMN B
(i) Neutral point	A. The region where magnetic fields cancel each other resulting in zero magnetic effect
(ii) Magnetization	B. The process of arranging magnetic domains in the same direction to make an object a magnet
(iii) Magnetic material	C. The invisible region around the earth that causes a freely suspended magnet to point North-south
(iv) Storage of magnets	D. Substance such as iron, nickel and cobalt that can be magnetized
(v) Earth's magnetic field	E. Keeping bar magnets in pairs with unlike poles together using soft iron keepers

LIST A	i	ii	iii	iv	v
LIST B					

SECTION B (70 marks)

3. During a school inspection, a student walks along a straight corridor whose length has been measured as **40m**. The student takes **25s** to move from one end of the corridor to the other without changing speed.

(a) From the information given, determine the physical quantities involved and their appropriate SI units. **(2marks)**

____

____

____

(b) Use the data to determine the student's rate of motion along the corridor. **(4marks)**

____

____

____

(c) Using the nature of the motion described, comment on the acceleration of the student. **(4marks)**

____

____

4. A student lifts a load vertically from the ground and places it on a table.

(a) Using this activity, explain the conditions under which work is considered to have been done. **(3marks)**

____

____

____

(b) Identify the form of energy acquired by the load after being lifted. **(3marks)**

____

____

____

(c) State two factors that would affect the amount of work done during the lifting process. **(4marks)**

____

____

5. A student is investigating why some objects float while others sink when placed in water.

(a) Define the following terms as used in his investigation:

(i) Up thrust **(1mark)**

____

____

(ii) Apparent weight **(1mark)**

____

____

(b) In his investigation some Principles and Laws were used.

(i) State Archimedes' Principle. **(2 marks)**

____

____

____

(ii) State one condition for sinking and floating of a body in a liquid. **(2mark)**

____

____

(c) A solid object weighs 50 N in air. When completely immersed in water, its apparent weight becomes 35 N.

Calculate:

(i) The upthrust acting on the object. **(2 mark)**

____

____

____

____

(ii) The relative density of the object. **(2 marks)**

____

____

____

____

6. A water tank is constructed at a height of 12 m above the ground to supply water to a school laboratory. Students observe that water flows faster from lower taps than from upper taps. They also use a hydraulic jack in the workshop to lift a car.

(a) Derive the expression for pressure in liquids at depth h, starting from the definition of pressure. **(2 marks)**

____

____

____

____

____

(b) From observation

(i) Explain why water flows with greater force from taps located lower down the building. **(2 marks)**

____

____

____

____

(ii) State two assumptions made when deriving the formula in part (a) **(2 mark)**

(c) Use the given information below:

(i) Calculate the pressure exerted by water at a depth of 5 m. **(2 marks)**

(ii) A hydraulic jack has a small piston of area $0.01m^2$ and a large piston of area $0.5m^2$. If a force of 200 N is applied on the small piston, calculate the force produced on the large piston. **(2 marks)**

7. A laboratory technician is measuring the external diameter of a cylindrical metal rod using a Vernier caliper. Before taking measurements, she checks the instrument carefully and ensures proper handling to obtain accurate results.

(a) Name and describe the function of any four main parts of a Vernier caliper. **(4 marks)**

(b) Explain two common sources of error when using a Vernier caliper. **(2 marks)**

(c) During measurement:

- The main scale reading is 2.4 cm
- The 6^{th} division on the Vernier scale coincides with a main scale mark
- The least count of the Vernier caliper is 0.01 cm

(i) Calculate the total diameter of the rod. **(2 marks)**

(ii) If a zero error of +0.02 cm is observed, determine the correct diameter. **(2 mark)**

8. (a) For each of the following, give a clear definition and an example:

(i) Elasticity **(2 marks)**

(ii) Cohesion **(2 marks)**

(iii) Adhesion **(2 marks)**

(b) (i) An object is placed 2cm from a plane mirror. If the object is moved by 1cm towards the mirror, what will be the new distance between the object and the image? **(2 marks)**

(ii) Two plane mirrors are placed at an angle of 60 to each other. When an object is placed between them, find the number of images formed due to repeated reflections. **(2 mark)**

9. (a) (i) State two effects of a force on an object. **(2 marks)**

(ii) A car of mass 800 kg is pushed with a force of 4000 N on a horizontal road. Calculate the acceleration of the car. **(2 marks)**

(iii) If the force is applied for 5 seconds, calculate the velocity acquired by the car, assuming it starts from rest. **(2 marks)**

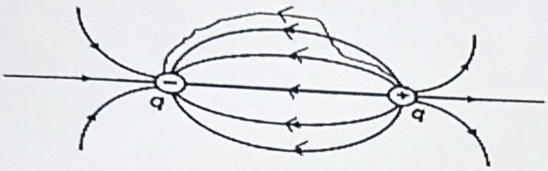
(b) A television set (TV) rated 40W is switched ON for 5 hours every day.

(i) How much electrical energy in KWh does this TV consume in 30 days? **(2Marks)**

(ii) If the price of a unit of electricity is Tsh.229.60, what is the total electricity cost of watching this TV in 30 days? **(2 marks)**

SECTION C (15 marks)

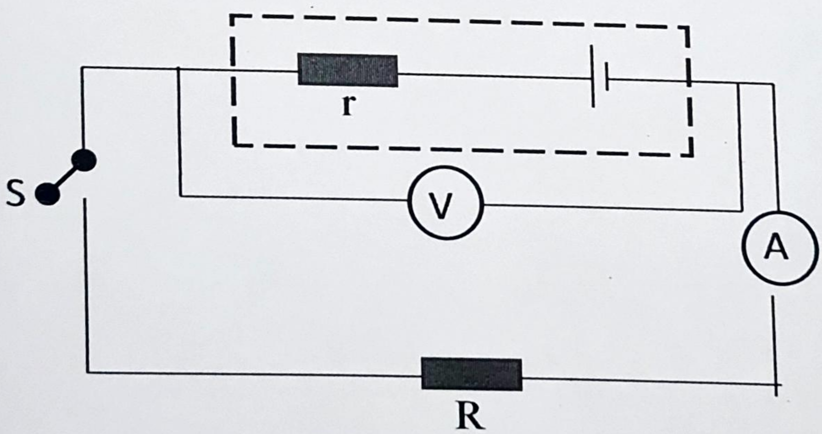
10. (a) After lesson on electric field the physics teacher left an exercise to the student. The drawing of one student was as shown here under:



Identify the mistake made by student in her drawing. (3marks)

(b) An aircraft flying for a long time through the atmosphere may acquire electric charge. Explain why passengers inside the aircraft are safe, but a ground crew member who opens the aircraft door immediately after landing may experience an electric shock. (3marks)

(c) Study the gure below and use the given instructions to answer the questions:



Switch condition	Ammeter reading (mA)	Voltmeter reading (V)
Open	0	9
Closed	250	7

(i) Why does the reading of the voltmeter drop when the switch is closed? **(2Marks)**

(ii) Calculate the unknown load resistance? **(4marks)**

(iii) Calculate the internal resistance **(3marks)**